

# Effect of Extreme Volume-Load Differences for a Single Unilateral Exercise During In-Season Resistance Training on Measures of Bilateral Strength, Power, and Speed in Collegiate American Football Players

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## Abstract

Crawley, K, Adams, KJ, DeBeliso, M, and Lawrence, MM. Effect of Extreme Volume-Load Differences for a Single Unilateral Exercise During In-Season Resistance Training on Measures of Bilateral Strength, Power, and Speed in Collegiate American Football Players. *J Strength Cond Res* XX(X): 000–000, 2023—This study examined the impact extreme volume-load differences for a single weekly exercise, when all other exercises' volume loads were similar, would have on American football performance variables after in-season resistance training (RT). Twenty male National Collegiate Athletic Association (NCAA) American footballers (18–23 years,  $98.4 \pm 19.3$  kg) were randomly assigned to an extreme high-volume low-intensity (EHVLI;  $n = 11$ ) group or a low-volume high-intensity (LVHI;  $n = 9$ ) group. Subjects performed the same evidence-based RT exercises and volume loads for 8 weeks thrice weekly, with the only differences being once weekly unilateral reverse dumbbell lunge (EHVLI) or unilateral Hatfield safety bar back squat (LVHI) exercises performed with different volume loads. Performance variables were assessed 1 week before (PRE) and after (POST) 8 weeks of RT. A 2-way analysis of variance with repeated measures and the Sidak post hoc test were used ( $p < 0.05$ ). Extreme high-volume low-intensity had no significant ( $p > 0.05$ ) PRE-to-POST RT changes in muscular strength in 1 repetition maximum ( $251.8 \pm 48.7$  to  $274.6 \pm 61.3$  kg) or power in vertical jump ( $79.2 \pm 8.8$  to  $78.2 \pm 10.8$  cm). Conversely, LVHI had significant ( $p < 0.05$ ) PRE-to-POST RT improvements in strength ( $249.2 \pm 54.4$  to  $284.1 \pm 55.0$  kg) and power ( $72.8 \pm 11.4$  to  $76.3 \pm 10.0$  cm). Furthermore, LVHI vs. EHVLI displayed significantly greater percentage difference increases from PRE values in muscular power ( $6.7\% \pm 7.2$  vs.  $-1.3\% \pm 6.0$ , respectively), with no significant differences between groups (LVHI vs. EHVLI) in muscular strength ( $8.8\% \pm 3.1$  vs.  $6.7\% \pm 8.0$ , respectively) and 10-yard acceleration ( $2.2\% \pm 5.6$  vs.  $3.2\% \pm 5.6$ , respectively). For in-season RT of strength and power in collegiate American football players, all exercises performed should use evidence-based volume loads to optimize adaptations because a single exercise performed with extreme volume load may limit muscular strength and power development.

**Key Words:** Hatfield, safety squat bar, explosive muscular power, acceleration

## Introduction

Annually, about 74,000 athletes compete in collegiate American football (from here on simply referred to as football) across all 3 divisions (I, II, and III) in the United States (NCAA, 2019 statistics). Football includes aspects of strength, power, and speed, and these variables can differentiate starters from nonstarters (11,13). Developing lower-body strength, power, and speed through carefully designed resistance training (RT) programs helps football athletes enhance general sport skill performance (e.g., sprinting and change of direction) and specific sport skill performance (e.g., explosive contact and position-specific abilities) and helps prevent injuries (7,12,14,30).

In-season programming for football should focus on strength, power, and rate of force development (8,14), but exactly how a strength coach designs and implements their program using scientific evidence remains poorly understood. To make matters

more challenging, it is easy for a strength coach to be overwhelmed with the vast number of options available when applying the acute program design variables. Therefore, from a research perspective, altering as few variables as possible should be performed per program to tease out how each contributes to specific adaptations. The 2 most commonly manipulated acute program design variables within a periodized program are volume and intensity (24). One common method to prescribe and track volume and intensity is volume load (volume as repetitions across all sets multiplied by intensity/load as body mass lifted in kilograms). In RT programs, it is well documented how prescribing specific volume loads across all exercises targets adaptations in specific neuromuscular characteristics (18,24,27). However, to our knowledge, it is unknown how one specific exercise's volume load (or volume and intensity combination) affects desired adaptations among all other prescribed exercises.

To this end, we observed a collegiate strength coach prescribing extremely high volume loads ( $\sim 18$ – $27\%$  1 repetition maximum [1RM] of 9–16 sets  $\times$  5–12 repetitions or 45–192 total repetitions) for a single exercise once weekly, which is not

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supported by evidence for training strength and power (24), when all other exercises in the weekly RT program were evidence based (24). This anecdotal exercise prescription provided a unique, real-world opportunity to determine whether prescribing a different exercise once a week after an evidence-based low-volume high-intensity (LVHI) volume load would alter performance variables compared with the extreme high-volume low-intensity (EHVLI) exercise's volume load when all other RT exercises and volume loads were identical.

Therefore, the purpose of this study was to examine the impact extreme volume-load differences for a single weekly exercise, when all other exercises and volume loads were similar, would have on football performance variables after in-season RT. We hypothesized that 8 weeks of in-season RT in Division I NCAA football players in the LVHI exercise group would result in significant increases in lower-body strength, power, and acceleration, whereas the EHVLI exercise group would result in maintenance or even decreases in the same variables. Findings from this investigation will help elucidate the role a single exercise's volume load has on in-season adaptations of strength, power, and speed in football athletes.

## Methods

### Experimental Approach to the Problem

This study followed a randomized experimental design (Figure 1). Subjects were assigned to experimental groups based on bilateral 1RM Hatfield safety bar back squat (HSBBS) to body mass (1RM/BM) lifted ratio and then randomly assigned to either group in the descending order from the highest to lowest 1RM/BM ratio. All subjects participated in the same strength and power-emphasized season-specific progressive periodized RT exercises and volume loads 3 times a week for 8 weeks, with the only difference being once weekly the EHVLI group performed the unilateral reverse dumbbell lunge exercise (9–16 sets  $\times$  5–12 repetitions at  $\sim$ 18–27% 1RM) and the LVHI group performed unilateral HSBBS exercise (4–6 sets  $\times$  3–10 repetitions at  $\sim$ 60–78% 1RM). All other variables (e.g., practices, game play, other RT exercises' volume load, nutrition, hydration, and recovery strategies, testing and training times, and strength coaches prescribing and implementing the RT program) remained constant for all subjects.

The overall 8-week RT program followed an undulating (nonlinear) periodization model with progressive overload and variation in volume and intensity occurring for training weeks 1, 2, 4, 5, 7, and 8 with training weeks 3 and 6 being deload weeks, following a 2:1 (2 weeks progressive overload and 1 week deloading) summated microcycle loading paradigm. In addition to the RT program exercises for 8 weeks, all subjects participated in practice 5 days a week and competitions once a week except bye weeks. Maximal lower-body strength, power, and acceleration were assessed 1 week before (PRE) and 1 week after (POST) the 8-week training program. In addition, during the deload weeks (training weeks 3 and 6), lower-body power and acceleration were measured.

### Subjects

Twenty-eight male NCAA Division I football athletes (18–23 years;  $98.4 \pm 19.3$  kg) at a university in the United States volunteered to participate, with 20 completing the study. The 8 subjects who did not complete the study chose to discontinue the

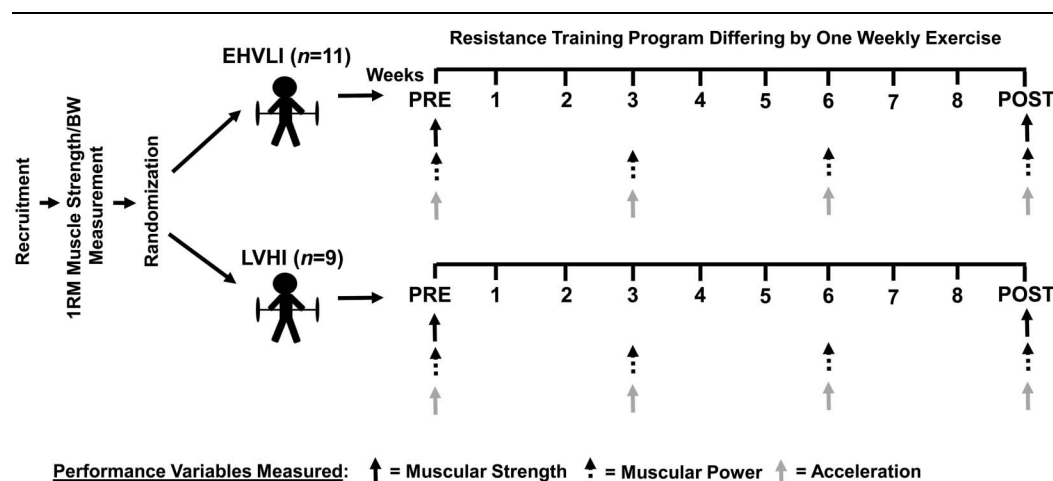
study for personal reasons, injury, or leaving the team. All subjects were on the active roster for the fall 2021 season and were classified as recreationally resistance trained (5) strength and power athletes, with between 1 and 5 years of RT experience. All subjects were nonstarters and were in a developmental group, such as NCAA redshirts. Permission to conduct the study was granted from the Southern Utah University Institutional Review Board (IRB) (#08-192021). Furthermore, IRB-approved informed consent was obtained from the athletes before the study initiation. Permission was also granted from all coaches.

### Procedures

This study took place over an 11-week period during the fall 2021 football season. All subjects performed the same season-specific progressive RT exercises and volume loads 3 times a week for 8 weeks (Table 1), with at least 48 hours between each session on Monday, Wednesday, and Friday of each week. The only difference in the groups was that once weekly (Monday), the EHVLI group ( $n = 11$ ) performed the unilateral reverse lunge exercise with dumbbells, and the LVHI group ( $n = 9$ ) performed the unilateral HSBBS exercise (Table 1) with various volume loads (Table 2).

Differences in strength and body mass size were accounted for across each group by randomly assigning subjects to either EHVLI or LVHI based on their bilateral 1RM/BM HSBBS. Specifically, the 2 athletes with the highest 1RM/BM bilateral HSBBS were randomly assigned, using a random number generator, to either the EHVLI or LVHI group. This pattern was followed from the highest 1RM/BM bilateral HSBBS to the lowest until all athletes were assigned to a group. The same NSCA Certified Strength and Conditioning Specialist (CSCS) strength coach designed and implemented all RT program exercises. The collegiate American football performance variables of lower-body maximal strength, power, and acceleration were assessed PRE and POST the 8-week training program. In addition, during the deload weeks (training weeks 3 and 6), lower-body power and acceleration were also measured. Two weeks before the initiation of the 8-week RT program (i.e., 1 week before the PRE measurements), all subjects were familiarized with all tests for the various performance variables. All subjects went through a standardized warm-up before any testing led by the same NSCA CSCS-certified strength coach, and the same strength coach also administered all tests. Verbal encouragement was also used for all trials across all tests for all testing days. All RT and testing were performed in the same training facility at the same time of the day. All subjects maintained normal dietary, hydration, and recovery practices as prescribed by the coaching and nutrition staff. On days where multiple different performance variables were tested, at least 5 minutes of rest between each different test was used (9).

**Lower-Body Muscular Strength 3RM Testing to Estimate 1RM.** Lower-body maximal strength was determined using a bilateral HSBBS following the published NSCA 1RM bilateral back squat protocol (10). Specifically, all strength assessments involved bilateral HSBBS with the weight on the back and the subjects' hands on handles attached to the rack in front of them for balance and not pulling themselves up (17). Subjects first performed a light warm-up set with 6–8 repetitions, followed by a 1 minute of rest. Then, subjects performed 2 more additional heavier warm-up sets increasing body mass  $\sim$ 10–20% for each set and performing less repetitions (i.e., 5–7 repetitions on the first set, 4–6 repetitions on



**Figure 1.** Overview of experimental design: NCAA Division I collegiate American football athletes performed the same season-specific progressive resistance training exercises for 8 weeks, with the only difference being once weekly the extreme high-volume low-intensity (EHVLI,  $n = 11$ ) group performed the unilateral reverse lunge exercise, whereas the low-volume high-intensity (LVHI,  $n = 9$ ) group performed the unilateral Hatfield safety bar back squat exercise. The collegiate American football performance variables of bilateral lower-body maximal muscular strength, explosive muscular power, and acceleration were measured 1 week before (PRE) and 1 week after (POST) the 8-week program. Explosive muscular power and acceleration were also measured on weeks 3 and 6 of the program.

the second set), with 2 minutes of rest per set. The subjects then performed attempts at a 3RM until a true 3RM was achieved. All athletes completed a 3RM in 3–5 attempts. The NSCA recognizes 1RM and 3RM as reliable measures of muscle strength (20). Published reliability coefficients ( $r \geq 0.90$  and  $ICC \geq 0.90$ ) confirm that 1RMs and 3RMs are extremely reliable measures (31). The actual 3RM lower-body strength value was then used to estimate a 1RM lower-body strength value using the following equation from Reynolds et al. (25):  $(1RM [kg] = 1.09703 (3RM \text{ body mass } [kg]) + 14.2546)$ . Reynolds et al. have reported that up to 5RM testing (and not 10RM or 20RM) yields highly valid ( $r \geq 0.96$ ) upper- and lower-body strength-estimated 1RM values compared with true 1RM strength, when using the developed predicted equation (25).

**Lower-Body Muscular Power.** Lower-body power was determined using a maximal countermovement vertical jump height

test from a jump mat contact system (Just Jump System; Probotics Inc., Huntsville, AL) following a published protocol (10). Briefly, after a standardized warm-up, subjects were instructed to place both feet on the jump mat and, without a preparatory or stutter step, perform a countermovement take-off by quickly flexing the knees and hips, moving the trunk forward and downward, and swinging the arms backward to then jump as high as possible reaching up with the dominant hand. For all trials, the same strength coach instructed all athletes to use the same countermovement take-off with consistent arm action and reaching as high as possible. Careful attention and instruction was also used for the landing technique, such that no exaggeration of hip flexion or knee flexion was allowed as to not inflate the countermovement vertical jump height measurement (21). Also, athletes had to take-off and land on the jump mat. The best of 3 trials was recorded based on the jump mat contact time, with 1-minute rest between trials. The NSCA recognizes jump mat contact systems

**Table 1**

**Resistance training exercises by day for training weeks 1–8 (exercises, sets  $\times$  reps).\***

| Day 1                                  | Day 2                                | Day 3                                 |
|----------------------------------------|--------------------------------------|---------------------------------------|
|                                        | Training weeks 1–4                   |                                       |
| EHVLI/LVHI exercises                   | Bench press 3–7 $\times$ 7–10        | Trap bar deadlift 4–7 $\times$ 4–12   |
| Plyometrics† 1–2 $\times$ 6–8          | Weighted pull-ups 3–7 $\times$ 5–10  | Cable goblet squat 4–7 $\times$ 10–12 |
| Part. back extensions 5–7 $\times$ 5–8 | DB In. bench press 3–7 $\times$ 7–10 | DB box step-up 4–7 $\times$ 6–12      |
|                                        | St. two-arm CR 3–7 $\times$ 7–15     | SL part. Ham. curl 4–7 $\times$ 5–8   |
|                                        | Hindu push-ups 3–7 $\times$ 7–15     |                                       |
|                                        | Training weeks 5–8                   |                                       |
| EHVLI/LVHI exercises                   | Bench press 6–8 $\times$ 1–6         | Barbell complex 5 $\times$ 2–4‡       |
| Plyometrics† 1–2 $\times$ 6–8          | Weighted pull-ups 6–8 $\times$ 1–6   | RDL 5 $\times$ 2–4                    |
| Part. back extensions 5–7 $\times$ 5–8 | DB In. bench press 1–5 $\times$ 8–14 | Trap bar deadlift 3–6 $\times$ 1–6    |
|                                        | St. two-arm CR 1–5 $\times$ 8–15     | Cable goblet squat 3–6 $\times$ 3–5   |
|                                        | Hindu push-ups 5 $\times$ 10         | DB box step-up 3–6 $\times$ 3–5       |
|                                        |                                      | Nordic Ham. Ecc. 2–5 $\times$ 4–10    |

\*EHVLI = extreme high-volume low-intensity; LVHI = low-volume high-intensity; Part. = partner; DB = dumbbell; St. = standing; CR = cable rows; SL = single leg; Ham. = hamstrings; RDL = Romanian deadlift; Ecc. = Eccentrics.

†Plyometrics included (6–8 exercises were performed each session): pogo jumps, squat jumps, rocket jumps, star jumps, lunge jumps, broad jumps, box jumps, and depth jumps.

‡Barbell complex included clean pull, hang clean, front squat, and push press.

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Table 2

Volume, intensity, weight lifted, volume load, and total work by groups.\*

| Training week                                                                        | 1                                         | 2                                         | 3                                      | 4                                         | 5                                         | 6                                      | 7                                         | 8                                         |
|--------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------|----------------------------------------|-------------------------------------------|-------------------------------------------|----------------------------------------|-------------------------------------------|-------------------------------------------|
| Extreme high-volume low-intensity (EHVLI), unilateral reverse lunge, $n = 11$        |                                           |                                           |                                        |                                           |                                           |                                        |                                           |                                           |
| Sets × Reps                                                                          | 15 × 6<br>@18%                            | 16 × 8<br>@18%                            | 12 × 5<br>@22%                         | 12 × 10<br>@23%                           | 15 × 12<br>@24%                           | 9 × 6<br>@24%                          | 15 × 12<br>@26%                           | 16 × 12<br>@27%                           |
| @ %1RM                                                                               |                                           |                                           |                                        |                                           |                                           |                                        |                                           |                                           |
| Wt. lifted                                                                           | 43.4 ± 7.9 (CI: 38.6–48.3)                | 48.6 ± 8.6 (43.4–59.4)                    | 53.5 ± 9.5 (47.9–59.4)                 | 55.9 ± 9.9 (50.0–62.0)                    | 56.1 ± 10.4 (49.8–62.4)                   | 60.8 ± 10.8<br>(54.4–67.4)             | 63.3 ± 10.7 (56.5–70.1)                   | 66.5 ± 11.8 (58.9–74.0)                   |
| VL                                                                                   | 3,905.9 ± 711.6<br>(3,475.8–4,336.2)      | 6,113.6 ± 1,395.7<br>(5,267.0–6,956.8)    | 3,118.3 ± 717.3<br>(2,684.8–3,551.8)   | 6,709.9 ± 1,189.0<br>(5,991.4–7,428.5)    | 10,097.1 ± 1,866.7<br>(8,969.2–11,225.0)  | 3,282.0 ± 581.6<br>(2,930.5–3,633.4)   | 11,387.3 ± 1,930.2<br>(10,160.9–12,613.8) | 12,759.7 ± 2,256.7<br>(11,325.8–14,193.6) |
| Total work                                                                           | 12,710.8 ± 1,957.4<br>(11,528.0–13,893.7) | 18,290.4 ± 3,736.0<br>(16,032.7–20,548.0) | 8,758.9 ± 1,738.0<br>(7,708.6–9,809.1) | 18,449.8 ± 2,828.1<br>(16,740.8–20,158.7) | 27,706.9 ± 4,525.7<br>(24,972.1–30,441.8) | 8,565.0 ± 1,309.3<br>(7,773.8–9,356.1) | 29,305.4 ± 4,303.7<br>(26,571.0–32,039.9) | 31,872.4 ± 4,653.8<br>(28,915.6–34,829.3) |
| Low-volume high-intensity (LVHI), unilateral Hatfield safety bar back squat, $n = 9$ |                                           |                                           |                                        |                                           |                                           |                                        |                                           |                                           |
| Sets × Reps                                                                          | 4 × 4<br>@60%                             | 6 × 4<br>@65%                             | 4 × 3<br>@68%                          | 5 × 6<br>@68%                             | 5 × 8<br>@67%                             | 4 × 3<br>@69%                          | 4 × 10<br>@75%                            | 5 × 8<br>@78%                             |
| @ %1RM                                                                               |                                           |                                           |                                        |                                           |                                           |                                        |                                           |                                           |
| Wt. lifted                                                                           | 137.1 ± 27.2† (CI: 119.7–154.3)           | 151.9 ± 31.8† (130.7–173.3)               | 149.5 ± 21.0† (135.3–163.3)            | 138.8 ± 31.0† (117.8–159.6)               | 154.1 ± 30.9† (133.2–174.8)               | 160.1 ± 33.7† (137.5–182.8)            | 166.7 ± 35.4† (141.3–191.9)               | 173.8 ± 37.0† (147.4–200.2)               |
| VL                                                                                   | 2,194.3 ± 435.2† (1,917.8–2,470.8)        | 3,645.6 ± 762.1† (3,133.7–4,157.6)        | 1,794.3 ± 251.4† (1,625.3–1,963.2)     | 4,165.3 ± 930.1† (3,540.5–4,790.1)        | 6,164.8 ± 1,236.5† (5,334.1–6,995.5)      | 1,990.1 ± 522.9† (1,638.8–2,341.4)     | 6,523.2 ± 1,420.2† (5,507.3–7,539.2)      | 8,235.8 ± 2,453.1† (6,493.9–9,977.8)      |
| Total work                                                                           | 3,710.7 ± 669.6† (3,285.2–4,136.1)        | 5,948.4 ± 1,100.8† (5,208.9–6,687.9)      | 2,945.7 ± 385.1† (2,686.9–3,204.4)     | 7,043.8 ± 1,360.5† (6,129.8–7,957.8)      | 10,002.8 ± 1,827.6† (8,775.0–11,230.6)    | 3,175.6 ± 730.4† (2,684.9–3,666.2)     | 10,292.0 ± 2,128.6† (8,769.3–11,814.7)    | 12,666.7 ± 3,106.3† (10,444.6–14,888.9)   |
| %TW                                                                                  | 29.2                                      | 32.5                                      | 33.6                                   | 38.2                                      | 36.1                                      | 37.1                                   | 35.1                                      | 39.7                                      |

\*Wt. = weight (in kg); VL = volume load (volume × body mass lifted in kg); total work (volume × external load without displacement in kg), TW = total work; %TW = LVHI percentage of EHVLI's total work; ANOVA = analysis of variance. Values are mean ± SD and 95% confidence intervals (CIs). A 2-way ANOVA (exercise × time, with repeated measures for time) was used. Post hoc comparisons were accomplished through a Sidak test, with significance set at  $p < 0.05$ .

†Significantly different from EHVLI for the same parameter. Percentage of 1RM (1 repetition maximum) values based on an individual's bilateral Hatfield safety bar back squat (HSBBS) 1RM.

(including the Just Jump System used herein) as valid and reliable measurements of maximal countermovement vertical jump height (9,16,21). Indeed, compared with the criterion-referenced 3-camera motion analysis system, the Just Jump System is a highly valid ( $r \geq 0.97$ ) measurement of countermovement vertical jump height (16). Furthermore, the Just Jump System has also been reported to be highly reliable ( $ICC \geq 0.90$ ) for measuring maximal countermovement vertical jump height (21). The best vertical jump height for each testing day was then used to estimate peak lower-body power output using the following prediction equation from Harman et al. (11): (peak power [watts] = 61.9 (jump height [cm]) + 36.0 (body mass [kg]) + 1,822). Harman et al. (11) reported the estimated lower-body peak power from vertical jump height using this equation to be highly valid ( $r = 0.88$ ) compared with the criterion-referenced force plate.

**Lower-Body Acceleration.** Lower-body acceleration was determined using a 10-yard straight-line sprint test with timing gates (Dashr System, Lincoln, NE) following a published protocol (10). Specifically, after a standardized warm-up, the subject was given an identification badge to scan in their unique identifier for each attempt with the timing gate system for their attempt. All attempts were performed on an indoor turf surface free of obstructions, with the sprinting course marked with a premeasured 10-yard interval from the start to finish timing gate. Each subject was instructed to start in a 2-point stance in a static position (i.e., no pre-emptive backward or forward movement) with their toe behind the starting timing laser line and that breaking the start timing laser gate would start the attempt. The subject was instructed to run in a straight line from a static start through the starting timing gate to the other timing laser gate placed exactly at 10 yards. When subjects passed the ending timing gate, the system would release an audible sound letting the subject know that they cleared the finish timing laser gate. The timing gates were all placed at ~76 cm (30 inches) high. The subjects maximally sprinted 10 yards and were given 30 yards for deceleration. Three trials were performed, with 2 minutes of rest between trials, and the best of 3 trials was recorded. The NSCA recognizes timing gates as a valid and reliable measurement of straight-line sprinting speed, including acceleration (9). Timing gates, including the Dashr system, have been shown to be highly valid ( $r \geq 0.97$ ) compared with the criterion-reference speed gun camera (21). Furthermore, Waldron et al. (32) reported timing gates to also be highly reliable (coefficient of variation  $\leq 1.5\%$ ) in measuring maximal straight-line sprinting.

**Unilateral Reverse Dumbbell Lunge (EHVLI) and Unilateral Hatfield Safety Bar Back Squat (LVHI) Exercises.** As described above, all subjects performed the same season-specific progressive RT exercises and volume loads 3 times a week for 8 weeks. The only difference between experimental groups was that once weekly the EHVLI group performed the unilateral reverse dumbbell lunge exercise with a volume load of 9–16 sets  $\times$  5–12 repetitions at ~18–27% 1RM, and the LVHI group performed the unilateral HSBBS exercise with a volume load of 4–6 sets  $\times$  3–10 repetitions at ~60–78% 1RM. The 1RM values for both groups were based on their 1RM from the PRE-bilateral HSBBS (i.e., the PRE maximal muscular strength measurement described above). For training weeks 1–8, the sets, repetitions, bilateral HSBBS percentage of 1RM used, body mass lifted, volume load, and total work differences between EHVLI and LVHI exercises are described in Table 2. Before any RT exercise, all subjects performed a standardized 10-minute dynamic warm-up prescribed by the same CSCS-certified strength coach.

The EHVLI unilateral reverse dumbbell lunge group is set up by the subject holding dumbbells in both hands in a standing position (Figure 2). The subject then takes one leg and steps back into a lunge position with their knee touching the ground or slightly above and then returns to the starting position. The subject then repeats this movement for the other leg, which is considered 1 repetition. For rest periods between sets, 1–3 minutes was used for this exercise.

The LVHI unilateral HSBBS is set up with a safety squat bar (Power Lift; Conner Athletic Products, Inc., Jefferson, IA) loaded with the prescribed weight, with the bar weighing ~31.82 kg (70 lbs). The safety squat bar as a training implement can create benefits for certain athletes as the squat becomes more upright through the torso placing less stress on the lower back and decreases compromising positions of the shoulder compared with a traditional barbell back squat (17,19). With the bar and weight on the subjects' back, the athlete placed their hands on handles attached to the rack in front of them for balance (i.e., not pulling themselves up) and assumed a staggered stance position with one foot in front of the other (Figure 3). The subjects then lower the trailing leg down to their knee touching the ground or slightly above and then standing up without moving their feet (Figure 3). The subjects repeated this movement for the other leg, which is considered 1 repetition. For rest periods between sets, 2–3 minutes was used for this exercise.



**Figure 2.** Overview of unilateral reverse lunge exercise movement performed by the extreme high-volume low-intensity (EHVLI) group ( $n = 11$ ).



**Figure 3.** Overview of unilateral Hatfield safety bar back squat (HSBBS) exercise movement performed by the low-volume high-intensity (LVHI) group ( $n = 9$ ).

### Statistical Analyses

A 2-way (experimental group  $\times$  time) analysis of variance (ANOVA) with repeated measures for time was used for any measurements with multiple time points. Post hoc comparisons were accomplished through a Sidak test for multiple comparisons, with statistical significance being set at  $p < 0.05$ . To compare the percentage difference across variables PRE-to-POST between the EHVLI and LVHI groups, an unpaired  $t$  test was used. Before any statistical analyses, the data were tested for normal distribution and equal variances to determine the appropriate statistical test. All statistical analyses and graphs were made using GraphPad Prism 9 (GraphPad, San Diego, CA). Data are presented as mean  $\pm$  SD.

A priori power analyses were conducted with G\*POWER 3.1.9.2 (Universitat Kiel, Germany) software. For between factors, within factors, or within-between interaction ANOVA with repeated measures, a statistical power of  $1 - \beta = 0.80$ ,  $\alpha = 0.05$ , and effect size of 0.50 as meaningful can be achieved with 18 subjects across 2 groups. The sample size in the current investigation was  $n = 20$  subjects across 2 groups. Also, as it has been argued (5) that  $p$ -value statistics may be less meaningful in real-world athletic settings where sample sizes may be small, and thus, effect sizes to show the magnitude of

interventional changes are also needed. We also report effect sizes. In particular, the effect size of each PRE-to-POST RT variable was calculated using Cohen's  $d$  using the following interpretations for the magnitude of  $<0.35$  as trivial,  $0.35$ – $0.80$  as small,  $0.80$ – $1.50$  as moderate, and  $>1.50$  as large, based on the training status classification of recreationally trained as previously described (5).

### Results

Twenty athletes participated in the study (EHVLI,  $n = 11$ ; LVHI,  $n = 9$ ). Overall body mass lifted (kg), volume load (kg), and total work (volume  $\times$  external load, kg), were significantly higher for the EHVLI group's unilateral reverse lunge exercise compared with the LVHI group's unilateral HSBBS exercise for all 8 weeks of the RT program (Table 2).

All bilateral performance PRE measurements were not significantly ( $p > 0.05$ ) different between EHVLI and LVHI groups (Table 3). Extreme high-volume low-intensity had no significant PRE-to-POST training changes in strength or power. By contrast, LVHI had significant ( $p < 0.05$ ) improvements from PRE-to-POST training in strength and power (Table 3). Extreme high-volume low-intensity and LVHI had

**Table 3**  
One week before (PRE) and after (POST) 8 wk of progressive resistance training measurements by group.\*

| Measurement                   | Extreme high-volume low-intensity (EHVLI), $n = 11$ |                                             |                |       | Low-volume high-intensity (LVHI), $n = 9$  |                                             |                |      |
|-------------------------------|-----------------------------------------------------|---------------------------------------------|----------------|-------|--------------------------------------------|---------------------------------------------|----------------|------|
|                               | PRE                                                 | POST                                        | % Diff.        | $d$   | PRE                                        | POST                                        | % Diff.        | $d$  |
| Actual AS (3RM kg)            | 223.6 $\pm$ 44.4 (CI: 191.9–255.4)                  | 244.4 $\pm$ 55.9 (CI: 201.5–287.4)          | 6.7 $\pm$ 8.0  | 0.42  | 221.2 $\pm$ 49.6 (CI: 183.1–259.3)         | 253.0 $\pm$ 50.2† (CI: 214.5–291.6)         | 8.8 $\pm$ 3.1  | 0.62 |
| Estimated AS (1RM kg)         | 251.8 $\pm$ 48.7 (CI: 217.0–286.7)                  | 274.6 $\pm$ 61.3 (CI: 227.5–321.8)          | 6.7 $\pm$ 8.0  | 0.42  | 249.2 $\pm$ 54.4 (CI: 207.4–291.0)         | 284.1 $\pm$ 55.0† (CI: 241.8–326.4)         | 8.8 $\pm$ 3.1  | 0.62 |
| Estimated RS (1RM/BM kg)      | 2.7 $\pm$ 0.5 (CI: 2.3–3.0)                         | 2.8 $\pm$ 0.6 (CI: 2.3–3.3)                 | 6.7 $\pm$ 8.0  | 0.20  | 2.6 $\pm$ 0.7 (CI: 2.2–3.1)                | 3.0 $\pm$ 0.7† (CI: 2.5–3.5)                | 8.8 $\pm$ 3.1  | 0.54 |
| Vertical jump height (cm)     | 79.2 $\pm$ 8.8 (CI: 73.3–85.1)                      | 78.3 $\pm$ 10.8 (CI: 71.0–85.5)             | –1.3 $\pm$ 6.0 | –0.10 | 72.8 $\pm$ 11.4 (CI: 64.7–81.0)            | 76.3 $\pm$ 10.0† (CI: 68.7–84.0)            | 6.7 $\pm$ 7.2‡ | 0.33 |
| Lower-body peak power (watts) | 10,259.3 $\pm$ 350.0 (CI: 10,024.1–10,494.3)        | 10,202.1 $\pm$ 392.3 (CI: 9,938.5–10,465.6) | –0.5 $\pm$ 3.1 | –0.16 | 9,689.1 $\pm$ 581.7 (CI: 9,273.0–10,105.2) | 9,989.3 $\pm$ 463.3† (CI: 9,633.2–10,345.4) | 2.9 $\pm$ 3.0‡ | 0.56 |
| Acceleration (s)              | 1.70 $\pm$ 0.10 (CI: 1.64–1.76)                     | 1.74 $\pm$ 0.08 (CI: 1.68–1.79)             | 3.2 $\pm$ 5.6  | 0.40  | 1.69 $\pm$ 0.12 (CI: 1.61–1.76)            | 1.74 $\pm$ 0.10 (CI: 1.67–1.82)             | 2.2 $\pm$ 5.6  | 0.52 |

\*ANOVA = analysis of variance;  $d$  = Cohen's  $d$  effect size; AS = absolute strength; RM = repetition maximum; RS = relative strength; Diff. = difference. Values are mean  $\pm$  SD and 95% confidence intervals (CIs). A 2-way ANOVA (exercise  $\times$  time, with repeated measures for time) was used. Post hoc comparisons were accomplished through a Sidak test, with significance set at  $p < 0.05$ .

†Significantly different from the PRE value within specific group.

‡Significantly different from EHVLI for same measurement.

no significant PRE-to-POST changes in acceleration (Table 3). Furthermore, LVHI compared with EHVLI resulted in significantly greater percentage change increases from PRE values in power measurements, with no differences between the groups in strength and acceleration (Table 3). No significant differences were observed for measurements of power and acceleration when comparing PRE or POST values with training week 3 or week 6 values for either EHVLI or LVHI (data not shown). Also, as the measured 3RM values resulted in the same outcomes as the estimated 1RM values (Table 3), we will refer to the estimated 1RM values alone.

## Discussion

This study investigated the impact extreme volume-load differences for a single weekly exercise, when all other exercises and volume loads were similar, would have on collegiate football performance variables after in-season progressive RT. Predictions before the study were that the LVHI group would increase in all tested performance variables (i.e., lower-body strength, power, and acceleration), whereas the EHVLI group would result in maintenance or decrease in the same performance variables after 8 weeks of RT. The results only support part of this hypothesis, as the LVHI group displayed significant increases in both lower-body strength (9%) and power (4–7%). Whereas the EHVLI group showed maintenance as indicated by the nonsignificant increase in strength (7%) and nonsignificant decreases in power (~1%). Furthermore, both LVHI and EHVLI displayed maintenance of acceleration as shown by the 2 and 3%, respectively, nonsignificant increases in acceleration times.

The current investigation adds to the literature by showing that a single exercise's volume load can influence training adaptations. In particular, the LVHI group performed all exercises, including the unilateral HSBBS, using evidence-based volume loads for training strength and power (24) and resulted in significant in-season improvements in each. On the other hand, the EHVLI group performed all other RT exercises using evidence-based volume loads for training strength and power, except the unilateral reverse lunge exercise that was performed using volume loads not empirically supported (24) and resulted in no significant changes in the same measurements as the LVHI group.

It is well documented that prescribing specific volume loads to train certain neuromuscular characteristics across all exercises for a given RT window can lead to desired adaptations (17,24,27). For example, Schoenfeld et al. (27) found in a meta-analysis of 21 RT studies that gains in 1RM strength favored high-intensity (>60% 1RM) training compared with low-intensity training (≤60% 1RM) across all exercises. Lopez et al. (18) also showed in a meta-analysis of 28 RT studies when intensity/load is prescribed in repetitions per set that moderate-intensity (9–15 RM or ~65–77% 1RM) and high-intensity (≤8RM or ~80+ % 1RM) training is superior for strength adaptations compared with low-intensity (>15 repetitions or <65% 1RM) training across all exercises, with high intensity being considered superior to moderate intensity, although not significantly. The current investigation adds to previous findings as the LVHI group (~60–78% bilateral HSBBS 1RM) had significant in-season improvements in strength and power, whereas the EHVLI group (~18–27% bilateral HSBBS 1RM) showed no significant changes.

Furthermore, the 60–78% bilateral HSBBS 1RM loads used for the unilateral HSBBS exercise in this investigation fall within previous findings examining bilateral and unilateral loads in the same individuals. Specifically, Ebben et al. (4) reported that the 6RM of the unilateral lunge exercise was ~67% of the bilateral back squat 6RM for the same subjects. Assuming that the LVHI group's unilateral HSBBS 1RM was ~67% or ~166.94 kg of their baseline bilateral HSBBS 1RM (249.16 kg, Table 3), the actual load lifted (Table 2) for training weeks 1–8 was between ~82 and 104% 1RM of the ~67% unilateral HSBBS 1RM estimate. Following the same assumption, the EHVLI group's reverse lunge 1RM at 67% of their bilateral HSBBS 1RM (251.82 kg, Table 2) would be ~168.72 kg, and their actual load lifted (Table 2) for weeks 1–8 was between ~26 and 39% 1RM of the ~67% unilateral reverse lunge 1RM estimate. Regardless of whether these assumptions are true, the current findings further support the meta-analytical findings of Schoenfeld et al. (27) and Lopez et al. (18) that moderate-intensity (~65–77% 1RM) and high-intensity (~80+ % 1RM) training is superior for strength adaptations compared with low-intensity (<60% 1RM) training.

With regard to the volume (repetitions and sets) needed for strength adaptations, there is much more agreement in the literature related to repetitions than sets. Indeed, in the meta-analysis (18) above, lower repetitions per set at high (≤8RM or ~80+ % 1RM) or moderate (9–15RM or ~65–77% 1RM) intensities are superior for strength development compared with higher volumes per set at low intensities (>15 repetitions of <65% 1RM). These findings are very similar to others (18,24,27) and provide strong consensus for how many repetitions with the appropriate intensity/load should be performed per set to train strength. However, there is much more debate in the literature regarding the amount of sets needed per exercise or muscle group to maximize strength adaptations (2,15,22,25), and inclusion of poor quality primary studies may be partially to blame for the equivocal results (23). In the most recent meta-analysis on this topic that included 9 high-quality RT studies on males alone, Ralston et al. (23) found that medium (5–9 sets) and high (≥10 sets) weekly training sets per exercise were better for improvements in strength than low (≤5 sets) weekly training sets per exercise when training at moderate to high intensities (~74–85% 1RM). In the current investigation, the LVHI group's unilateral HSBBS exercise was performed using 4–6 sets per week with a weekly set mean across all 8 training weeks of 4.6 sets. The EHVLI group's unilateral reverse lunge exercise was performed using 9–16 sets per week with a weekly set mean across all 8 training weeks of 13.8 sets. In addition, the other exercises targeting the same muscle groups as the unilateral HSBBS and reverse lunge (Table 1) for both LVHI and EHVLI fell between 4 and 7 sets per exercise weekly. The current in-season investigation in football players does not support the previous Ralston meta-analysis findings because the EHVLI group (13.8 mean weekly sets per exercise) had inferior strength improvements compared with the LVHI group (4.6 mean weekly sets per exercise). However, the current investigation findings and the aforementioned strong evidence on the importance of moderate to high intensity for optimizing muscular strength gains (18,27) suggest that exercise intensity may contribute more to strength development than the set volume, although multiple sets are needed.

The current investigation findings that one exercise performed with an extreme volume load can negatively influence strength

and power adaptations were our expectation. The EHVLI's group volume load (9–16 sets  $\times$  5–12 repetitions at  $\sim$ 26–39% 1RM using the 1RM estimate above) does not follow the recommendations for RT strength or power but more closely resembles the recommendations (24) for RT local muscular endurance (40–60% 1RM,  $>15$  repetitions). Although the EHVLI group did not result in significant increases or decreases in strength or power, there are clear differences between the RT improvements seen with the LVHI group for the same performance variables. The nonsignificant  $\sim$ 7% increase or maintenance in strength seen in the EHVLI group after short-term RT is likely explained by the secondary exercises (e.g., trap bar deadlift, cable goblet squat, dumbbell box step-up, and Romanian deadlift) that targeted the same muscle groups as the bilateral HSBBS used for strength assessment. The other exercises were performed using recommended (24) volume loads for training strength and power and therefore likely helped the EHVLI athletes maintain in-season strength. However, the LVHI group had significant improvements in strength and power and performed the same secondary exercises as the EHVLI group, further supporting that a single exercise's volume load can negatively influence overall training gains for strength and power in-season in strength and power athletes. Moreover, the lack of significant adaptation in strength and power seen in the EHVLI group, compared with the LVHI group, is concerning because of the lack of improvement in a developing population (i.e., less RT experience/age) when exposed to a relatively new training experience.

It might be argued that the unilateral HSBBS group saw improvements in bilateral HSBBS 1RM because the unilateral variation was very similar to the bilateral HSBBS movement, and therefore, the training principle of specificity would suggest that the outcome was obvious (i.e., specific adaptations to imposed demands). Yet, when looking at power improvements that were measured on a different exercise movement (i.e., countermovement vertical jump), there are clearly differences between the LVHI and EHVLI groups, suggesting that the specificity of the exercise itself had less influence on the training improvements than did the volume-load differences. In fact, the total work (Table 2) performed by the EHVLI group for the weekly unilateral reverse dumbbell lunge exercise was  $\sim$ 60–70% greater than the total work performed by the LVHI for their weekly unilateral HSBBS exercise. Also, the EHVLI group's extreme volume-load muscular endurance type exercise was performed in the same training session as plyometric exercises (Table 1). Taken together, our current findings along with others (3,28) suggest that training muscular endurance and strength or power in the same RT window are potentially incompatible and therefore should be cautioned.

There is some evidence to suggest that aerobic endurance (not muscular endurance) volume can negatively influence muscular power. Indeed, Chambers et al. (3) found that power (vertical jump height) was reduced immediately and up to 18 days after a 90 km ultramarathon. Furthermore, a more recent meta-analysis of 43 RT studies (28) examining concurrent aerobic and RT (concurrent training) found that strength development was not affected by concurrent training. However, the same study found that power development was hindered by concurrent training, and this hindrance was more pronounced when the concurrent training was performed in the same session than if the concurrent training was separated by at least 3 hours (28). In the current investigation, the EHVLI group's extreme volume-load muscular endurance type exercise was performed in the same session as plyometric exercises weekly (Table 1). Taken together, our current findings along with others (3,28) may suggest that training

muscular endurance and strength or power in the same RT window are potentially incompatible and therefore should be cautioned. This is especially true in-season training in strength and power athletes when strength and power improvements are the main training goals.

Outside of expert opinion (8,12,14), few RT studies on collegiate football have been undertaken (12–14), with even fewer focusing exclusively on in-season training (13). The current investigation adds another in-season RT study for collegiate football and supports a mixed methods training approach (7) to improve strength and power. Using progressive nonlinear periodized RT with a combination of exercises focusing on various parts of the force-velocity curve continuum, including strength (e.g., unilateral HSBBS and trap bar deadlift), power (e.g., Olympic weightlifting barbell complexes), and ballistic exercises and plyometrics, resulted in significant increases in both lower-body strength (9%) and power ( $\sim$ 3–7%) in the LVHI group alone. To our knowledge, in the only other in-season RT study on collegiate football players, Hoffman et al. (13) examined 53 male NCAA Division III football players for improvements in upper- and lower-body strength as well as body composition changes after 12 weeks of twice weekly in-season RT. The only weekly exercises performed were the power clean, back squat, push press, and bench press for 3 sets of 3–6 and 6–8 repetitions, for exercises focusing on power and strength, respectively. The authors found that lower-body strength as measured by bilateral back squat 1RM significantly increased ( $\sim$ 5% or  $\sim$ 8.3 kg) after training with no significant changes, or maintenance, in upper-body strength or body composition. The authors further examined the variables of training volume, preseason strength level, year in school, and training intensity to see which, if any, predicted the strength improvements. The only variable to predict the strength improvements in-season was training intensity, with athletes who trained  $\geq$ 80% 1RM demonstrating greater strength gains than those who trained  $<$ 80% 1RM (13). The current investigation supports the training intensity findings of Hoffman et al. (13), as the estimated %1RM loads for the LVHI group's unilateral HSBBS exercise was  $>$ 80% 1RM ( $\sim$ 82–104% 1RM), which resulted in significant in-season gains in strength and power, whereas the EHVLI group's unilateral reverse lunge exercise ( $<$ 40% 1RM) did not.

An unexpected finding in the current investigation was that 8 weeks of progressive in-season RT resulted in maintenance of acceleration in both the LVHI and EHVLI groups. This was determined by the nonsignificant 2 and 3% PRE-to-POST increases in 10-yard sprint times in the LVHI and EHVLI groups, respectively. These findings suggest that the acceleration-specific training or sport-specific training stimulus was not sufficient to cause improvements and therefore could be better addressed in future programming. Indeed, others examining strength and power athletes from other sports have shown significant improvements in sprinting performance in-season (6,29). It is possible that the preseason measurements of sprinting (PRE, Table 3) in the current investigation (after off-season training with limited sport practice/competition stress) were the fastest sprint times the athletes would have throughout the training year, and the POST measurements of sprinting following the competitive season were influenced mainly by the overall training stress of the sport practices and competitions without proper fatigue management.

Another novel aspect of this study is that it provides, to our knowledge, the first evidence for the efficacy of any HSBBS variation (i.e., upper-extremity-assisted safety bar back squat

exercises) during RT. Anecdotally, the HSBBS, bilaterally and unilaterally, is becoming more and more popular for strength and conditioning professionals (1). However, to our knowledge, the empirical evidence to support the usage of the HSBBS squatting variation remains nonexistent (17). That said, in related work, Meldrum and DeBeliso (19) did find that squat training with a traditional Olympic barbell or safety squat bar can both result in similar improvements in lower-body strength and power in strength and power athletes.

With regard to possible limitations, this was a short-term (8-week) training study representing a typical strength and conditioning training window for collegiate football. However, longer term training may reveal different results. That said, because this was a real-world, university-based, competitive training study, it had limitations in duration, sport practice stress/volume, and athletes' academic load. Also, subjects were developing nonstarters who did not have the added physiological and psychological stress of playing meaningful minutes in competitions. Thus, the in-season RT results may be different in starters who must deal with the added recovery needed after competitions.

In conclusion, the results demonstrate the negative impact a single exercise's volume load can have on strength and power gains during collegiate football in-season RT when an extreme volume load more closely resembling recommendations for training muscular endurance is used. Furthermore, the results provide the first evidence demonstrating the efficacy of any HSBBS variation (i.e., upper-extremity-supported safety bar back squat), including the unilateral HSBBS variation, that can be used as a weekly exercise during RT to develop strength and power.

### Practical Applications

Resistance training program design is complex because of a vast number of acute program variable combinations available to the strength and conditioning coach. This study demonstrates that when the training goals are to optimize strength and power, all exercises prescribed during the training window should follow volume-load recommendations for strength and power, and prescribing a single weekly exercise using extreme volume-load-like recommendations for muscular endurance is not recommended. These results highlight the point that subtle variations in RT programming may influence training outcomes in unexpected ways over time (i.e., positive or negative). Results also support a mixed methods training approach to improve strength and power during the season in collegiate football players. Using progressive nonlinear periodized RT with a combination of exercises focusing on various parts of the force-velocity curve continuum, including strength (e.g., unilateral HSBBS and trap bar deadlift), power (e.g., Olympic weightlifting barbell complexes), and ballistic exercises and plyometrics may help optimize strength and power.

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